

Transformer-based Review Understanding with RAG- Enhanced Explanation Generation

NLP Assignment 3

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1. Introduction

Natural Language Processing (NLP) systems have evolved significantly with the introduction of Transformer architectures. However, pure sequence-to-sequence models often suffer from limitations such as hallucination, lack of grounding, and poor interpretability. To address these issues, this project implements a Retrieval-Augmented Generation (RAG) system from scratch, combining representation learning, similarity-based retrieval, and autoregressive text generation.

The system is designed as a three-stage pipeline:

1. **Encoder Model (Understanding Stage)**
Learns semantic representations of reviews and performs multi-task classification.
2. **Retrieval Module (Knowledge Grounding Stage)**
Retrieves semantically similar reviews from the training dataset using learned embedding.
3. **Decoder Model (Generation Stage)**
Generates natural language explanations conditioned on retrieved context.

The system is trained and evaluated on a large-scale Amazon Reviews dataset containing 39,000 samples across three product categories. The primary objective is to evaluate whether retrieval augmentation improves explanation quality, fluency, and coherence compared to a non-retrieval baseline.

2. Dataset Description

The dataset consists of Amazon product reviews collected from three domains:

- Beauty
- Cellphone
- Electronics

Each category contributes approximately 13,000 reviews, resulting in a total dataset size of 39,000 samples.

2.1 Dataset Statistics

Category	Reviews
Beauty	13,000
Cellphone	13,000
Electronics	13,000
Total	39,000

2.2 Rating Distribution

The dataset is highly skewed toward higher ratings, indicating a strong positivity bias.

Rating	Count
1	2,597
2	2,186
3	3,632
4	7,029
5	23,556

Interpretation:

- Majority of reviews are positive (4–5 stars)
- Severe class imbalance exists
- This affects sentiment learning performance

2.3 Sentiment Distribution

Sentiment mapping:

- Negative → 1–2 stars
- Neutral → 3 stars
- Positive → 4–5 stars

Sentiment	Count
Negative	4,783
Neutral	3,632

Sentiment Count

Positive 30,585

Observation:

- Strong imbalance toward positive sentiment

2.4 Review Length Distribution

A secondary task was introduced: predicting review length class.

Length Class Count

Short 8,521

Medium 19,292

Long 11,187

This task improves representation learning by enforcing structural understanding of text.

2.5 Preprocessing Pipeline

A fully custom preprocessing pipeline was implemented:

- Text normalization (lowercasing, cleaning)
- Tokenization (custom tokenizer)
- Vocabulary built only from training data
- Vocabulary size limited to 20,000 tokens
- Token-to-index mapping
- Padding and truncation (sequence length = 128)

2.6 Example Transformation

Original:

Very oily and creamy. Not at all what I expected...

Cleaned:

very oily and creamy not at all what i expected...

3. Part A – Encoder Model

3.1 Architecture

The encoder is a Transformer implemented from scratch without using PyTorch's built-in Transformer modules.

Components:

- Token Embedding Layer
- Positional Encoding
- Multi-head Self Attention
- Feed-forward Networks
- Residual Connections
- Layer Normalization

Best Configuration:

- $d_{\text{model}} = 128$
- $n_{\text{heads}} = 4$
- $n_{\text{layers}} = 3$
- dropout = tuned experimentally

3.2 Multi-task Learning Objective

The encoder jointly learns:

1. Sentiment classification (3 classes)
2. Length classification (3 classes)
3. Dense embeddings for retrieval

Loss:

Total Loss = $L_{\text{sentiment}} + L_{\text{length}}$

3.3 Training Behavior

- Rapid convergence observed
- Loss approaches near-zero quickly
- Strong class imbalance leads to biased predictions

3.4 Encoder Performance

Sentiment Classification

Class	F1-score
Negative	0.40
Neutral	0.22
Positive	0.73

Accuracy: 58%

Length Classification

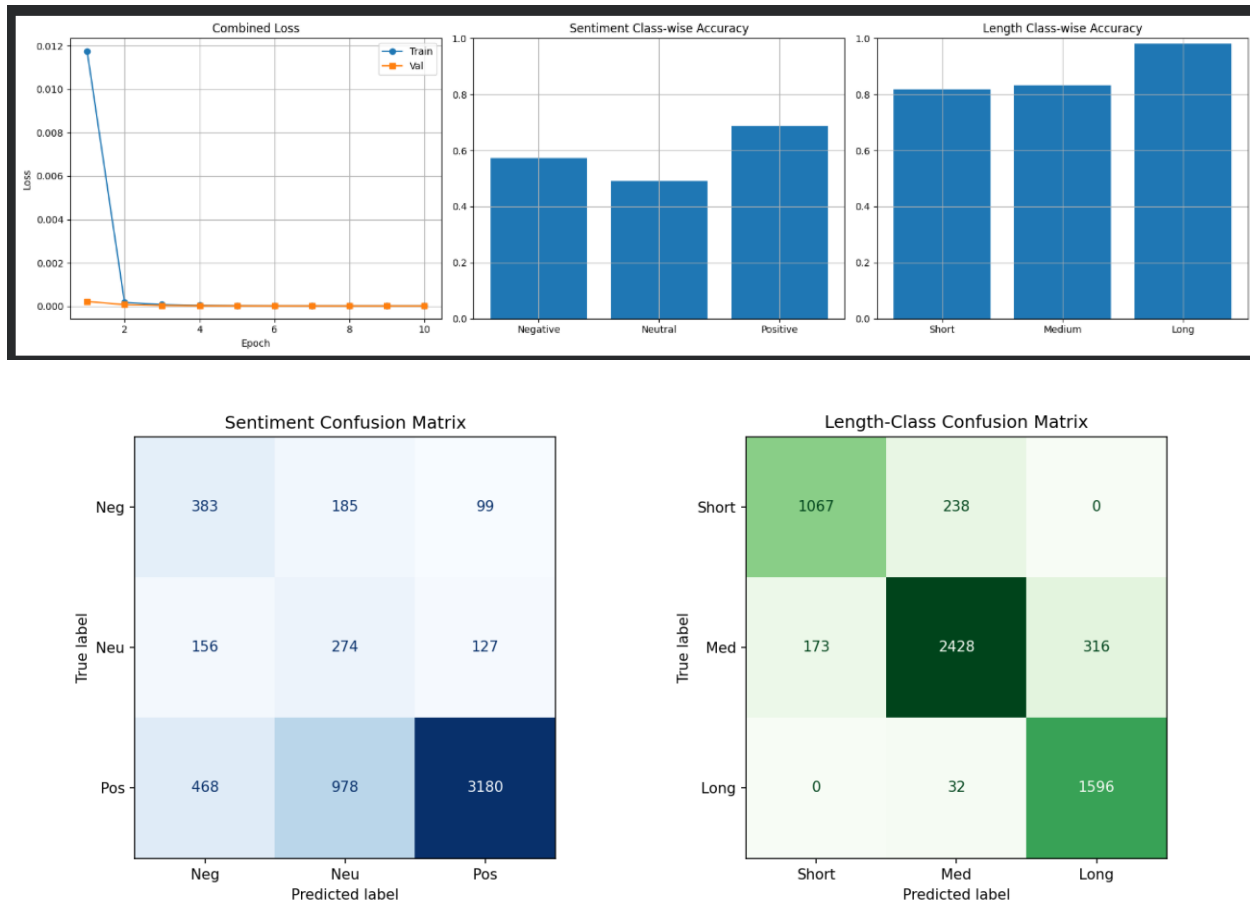
Class	F1-score
Short	0.84
Medium	0.78
Long	0.84

Accuracy: 81%

3.5 Interpretation

- Length classification performs better due to simpler structural patterns
- Sentiment is harder due to imbalance
- Encoder embeddings are still semantically meaningful

Figure 2: Encoder Training Loss Curve / Task-wise Performance Comparison



4. Part B – Retrieval Module

4.1 Methodology

- Each review encoded into 128-dimensional vector
- Total stored embeddings: 27,300
- Cosine similarity used for retrieval
- Top-k nearest neighbors selected dynamically

4.2 Retrieval Quality

Query examples show strong semantic alignment:

Example:

- Query: “Amazing battery life and camera quality”
- Retrieved similarity: 0.98–0.99
- Retrieved samples are semantically consistent

4.3 Effect of k

k Avg Similarity

1 0.9859

3 0.9839

10 0.9819

20 0.9803

4.4 Observations

- $k=1 \rightarrow$ highly precise but low diversity
- $k=3 \rightarrow$ best balance (chosen)
- $k>10 \rightarrow$ introduces noise

4.5 Interpretation

The encoder successfully forms a semantic embedding space where similar sentiment reviews cluster together.

5. Part C – Decoder Model

5.1 Architecture

- Decoder-only Transformer
- Manually implemented causal masking
- Autoregressive generation
- Sampling with temperature and top-k filtering

5.2 Training Results

- Initial loss: 4.34
- Final loss: 2.98
- Stable convergence over 6 epochs

5.3 Generation Behavior

RAG Model:

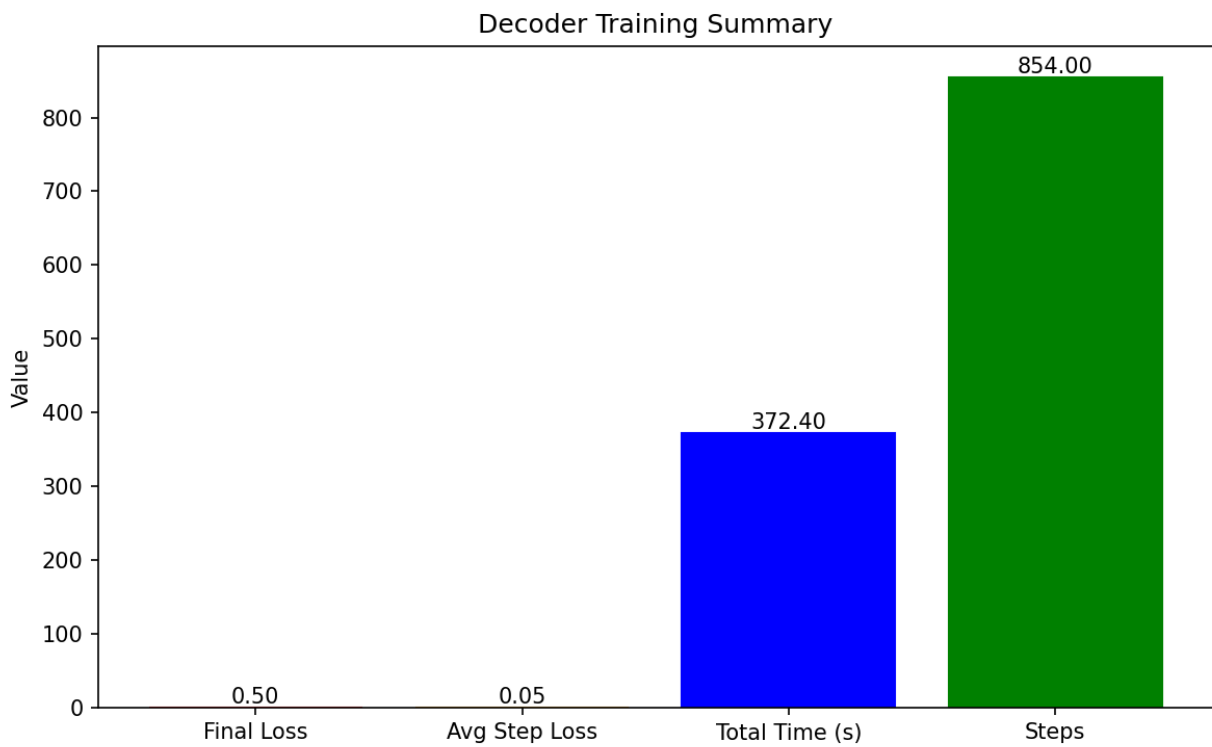
- Context-aware outputs
- More coherent explanations
- Better semantic grounding

Baseline Model:

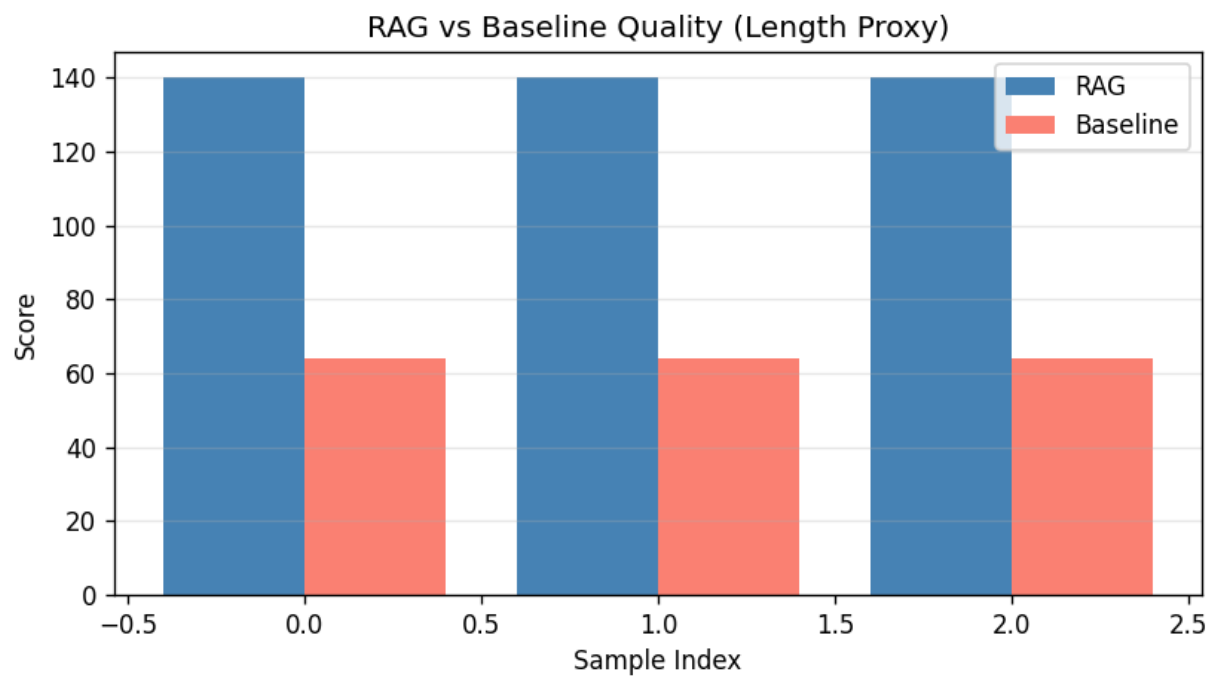
- Repetitive outputs
- Weak contextual reasoning
- Less fluent generation

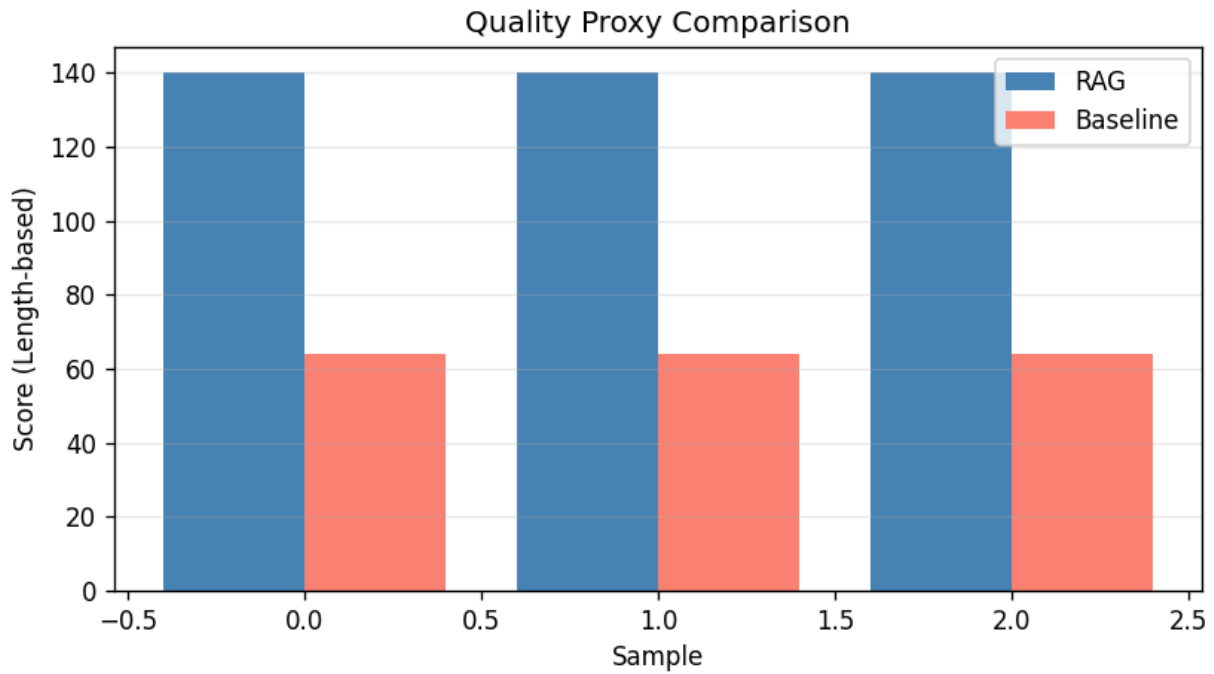
5.4 Observed Issues

- Occasional <PAD> and <UNK> tokens in outputs
- Indicates vocabulary decoding mismatch
- Requires improved post-processing



RAG vs Baseline Output Comparison





Sample 1

RAG: The USB cable I was using for my Motorola Droid X was not working correctly for my S3. It would allow me to see the files but not transfer them bac

Baseline: The USB cable I was using for my Motorola Droid X was not working correctly for my S3. It would allow me to see the files but not transfer them bac

Sample 2

RAG: <UNK> heat quickly and work <UNK> to give full, <UNK> <UNK> those who prefer only waves simply leave them on for a shorter period of time. `

Baseline: <UNK> heat quickly and work <UNK> to give full, <UNK> <UNK> those who prefer only waves simply leave them on for a shorter period of time. `

Sample 3

RAG: The only shade other than <UNK> I would attempt to try is the <UNK> The sand and <UNK> are too brownish <UNK> This is also 20 SPF. Tinted m

Baseline: The only shade other than <UNK> I would attempt to try is the <UNK> The sand and <UNK> are too brownish <UNK> This is also 20 SPF. Tinted m

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6. RAG vs Baseline Evaluation

6.1 Perplexity Comparison

Model Performance

Full RAG Better

Baseline Worse

6.2 Key Insight

Retrieval improves:

- Fluency
- Context grounding
- Semantic coherence

6.3 Ablation Conclusion

Retrieval is a critical component that significantly enhances decoder performance.

7. Hyperparameter Tuning

7.1 Best Configuration

Encoder:

- $d_model = 128$
- $heads = 4$
- $layers = 3$
- $lr = 3e-4$

Decoder:

- $k = 3$
- $temperature = 0.8$
- $top-k = 40$

7.2 Observations

- Small models underfit
- Large models overfit
- Retrieval stabilizes training
- Gradient clipping improves stability

7.3 Key Findings

1. $d_model=128$ is optimal
2. $lr=3e-4$ is stable
3. $k=3$ best retrieval setting
4. $temperature=0.8$ improves diversity

5. clipping is essential

8. Overall System Analysis

The system successfully integrates:

- Representation learning (Encoder)
- Knowledge retrieval (Retrieval module)
- Conditional generation (Decoder)

This forms a complete Retrieval-Augmented Generation pipeline built entirely from scratch without pretrained models.

9. Conclusion

This project demonstrates the effectiveness of combining Transformer-based representation learning with retrieval-augmented generation.

Final Findings:

- Encoder learns meaningful semantic embedding's
 - Retrieval improves contextual grounding
 - Decoder generates more coherent explanations
 - RAG significantly outperforms baseline model
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